

LLMs Powered by Retrieval-Augmented Generation (RAG)

THAÍS DE ARAÚJO DE MEDEIROS

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WHO AM I

THAÍS MEDEIROS

Member of the **Conect2ai**
Research Group | Projects:
CNPq and **Rota2030**



**Technician in
Textiles**
IFRN



**Bachelor of Science
and Technology**
UFRN



Master's Degree and PhD Candidate
Graduate Program in Electrical and
Computer Engineering
PPgEEC | UFRN



Computer Engineer
UFRN

AGENDA



Introduction



System Building



Architecture & Key
Components



Hands-On

Which Brazilian actress won the Golden Globe in 2025?



I don't know which Brazilian actress won the Golden Globe in 2025 because my training data goes only up to **June 2024**. Any events that happened after that date, including the 2025 ceremony, are outside of my knowledge. Without searching the web, I cannot provide a confirmed answer.



Which Brazilian actress won the Golden Globe in 2025?

Context:

Brazil woke up to a historic achievement: Fernanda Torres became the first Brazilian actress to win the Golden Globe in 2025. She was awarded for her performance in *I'm Still Here (Ainda Estou Aqui)*, Globoplay's first original film, in the category of Best Actress in a Motion Picture – Drama.

At the ceremony in Hollywood, Walter Salles celebrated the recognition and highlighted the film's impact on Brazilian audiences. Based on the book by Marcelo Rubens Paiva, the feature film portrays the struggle of Eunice Paiva to find her husband, Rubens Paiva, who disappeared during the military dictatorship.



Fernanda Torres won the Golden Globe in 2025, becoming the first Brazilian actress to receive the award. She earned it for her performance in *I'm Still Here (Ainda Estou Aqui)*, Globoplay's first original film, in the category of Best Actress in a Motion Picture – Drama.



Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis^{†‡}, Ethan Perez^{*},

Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

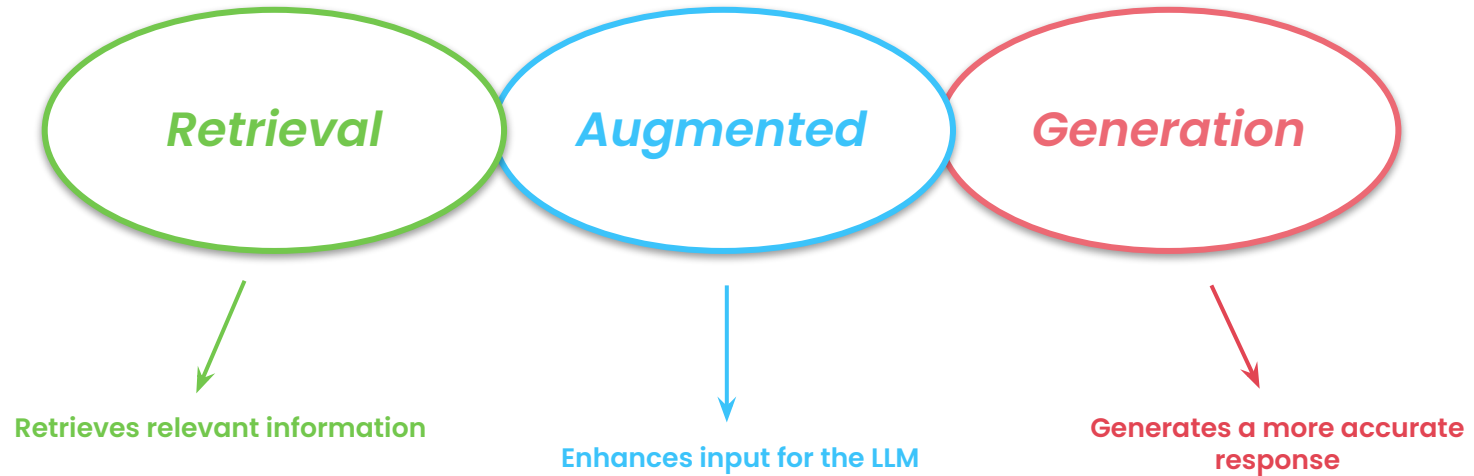
Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

[†]Facebook AI Research; [‡]University College London; ^{*}New York University;
plewis@fb.com

Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. Pre-trained models with a differentiable access mechanism to explicit non-parametric memory have so far been only investigated for extractive downstream tasks. We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) — models which combine pre-trained parametric and non-parametric memory for language generation. We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We compare two RAG formulations, one which conditions on the same retrieved passages across the whole generated sequence, and another which can use different passages per token. We fine-tune and evaluate our models on a wide range of knowledge-intensive NLP tasks and set the state of the art on three open domain QA tasks, outperforming parametric seq2seq models and task-specific retrieve-and-extract architectures. For language generation tasks, we find that RAG models generate more specific, diverse and factual language than a state-of-the-art parametric-only seq2seq baseline.

CONCEPT



HOW TO BUILD?

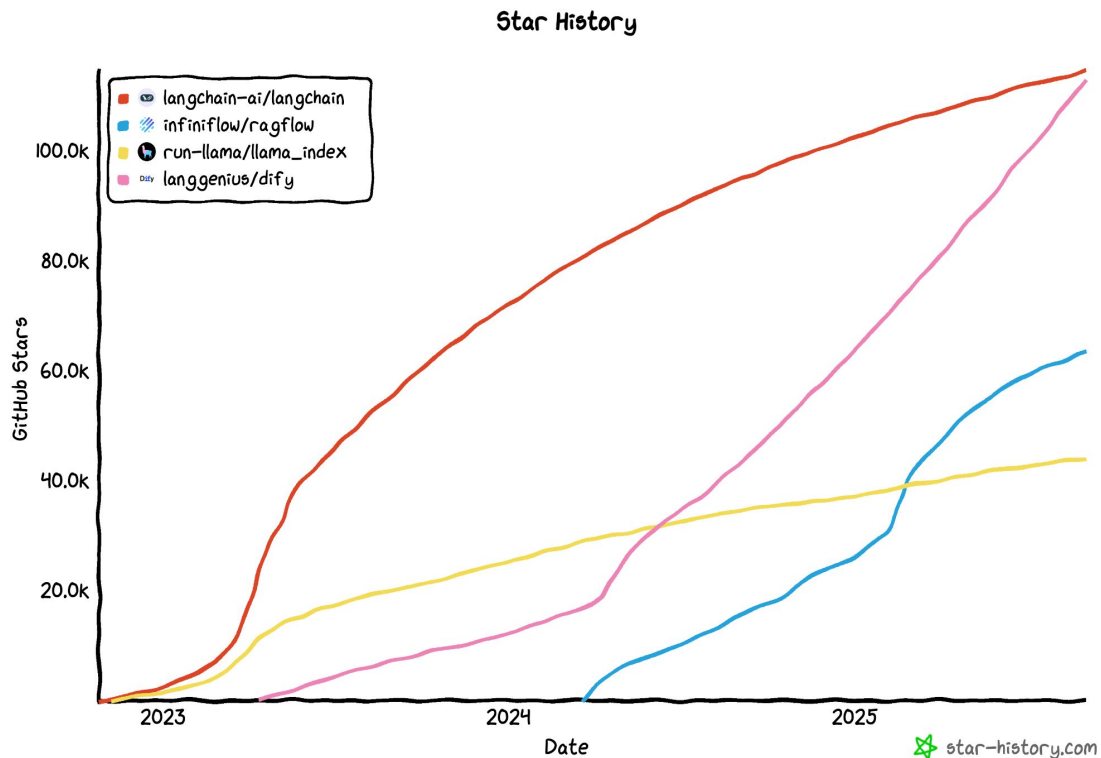


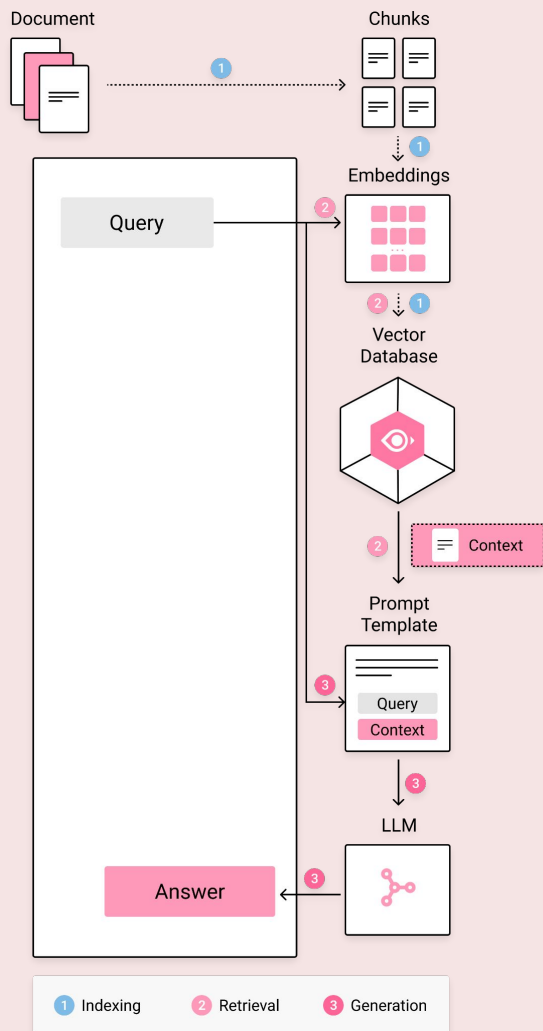
LlamaIndex



LangChain

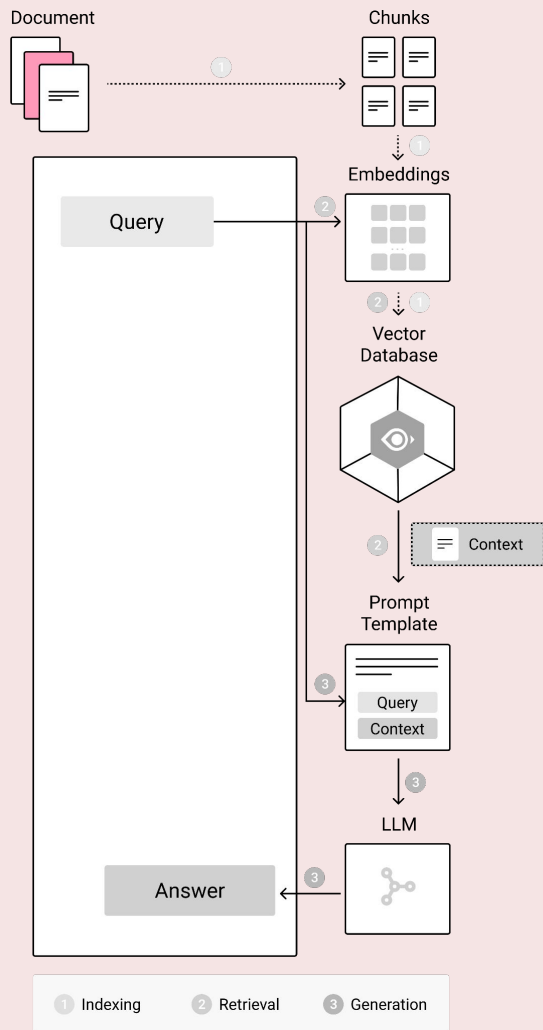
HOW TO BUILD?





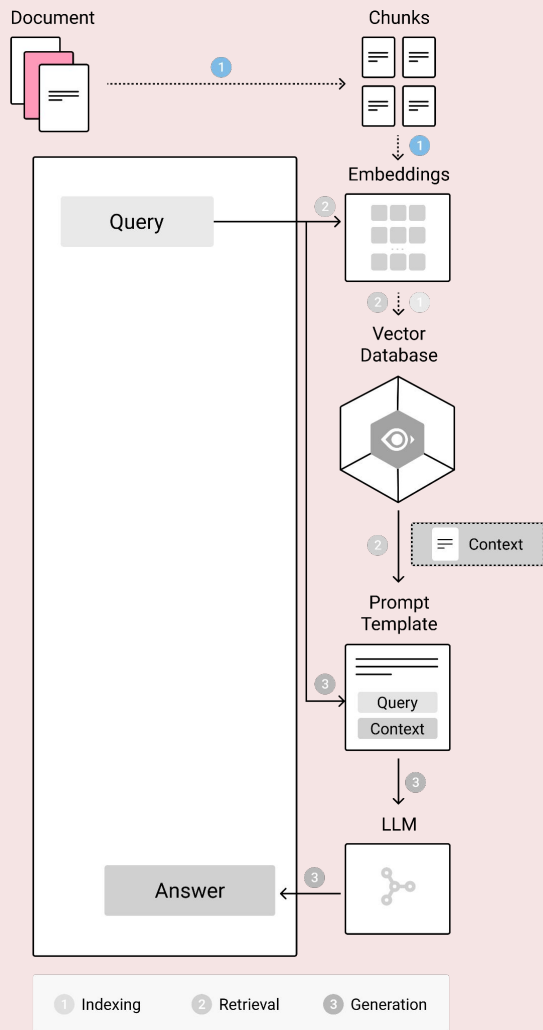
A TYPICAL RAG SETUP



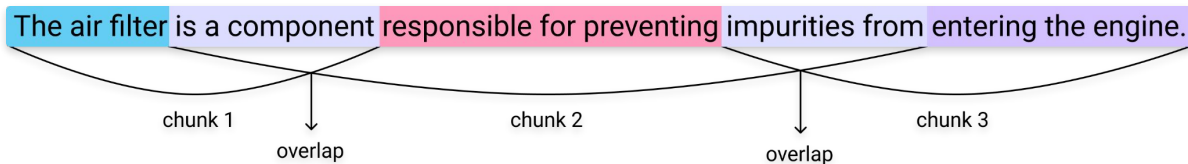


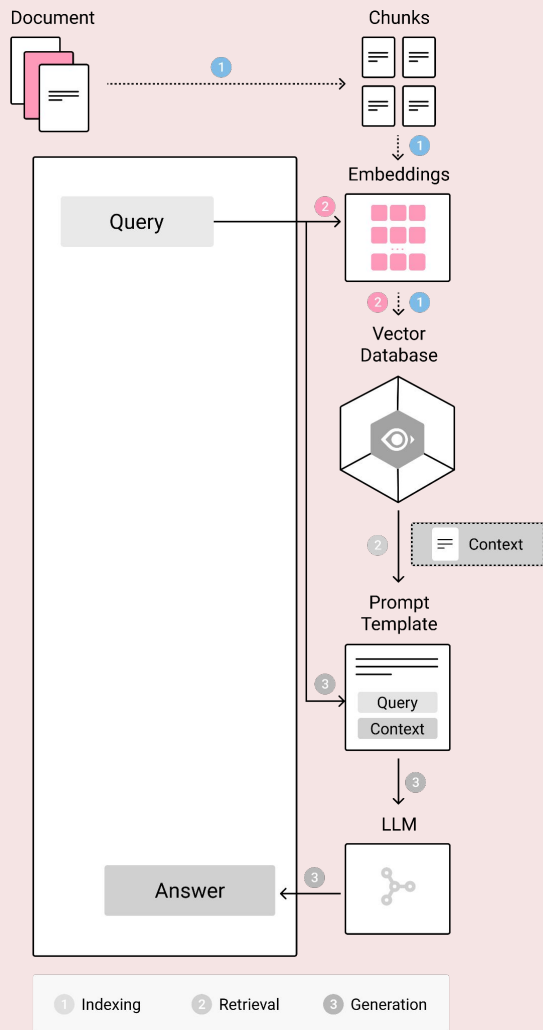
TYPES OF DOCUMENTS



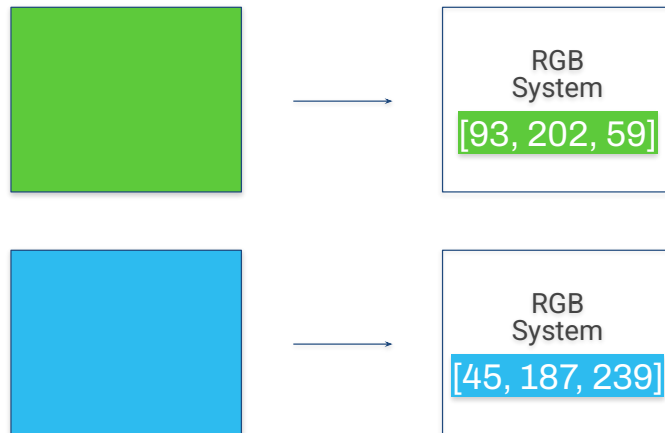


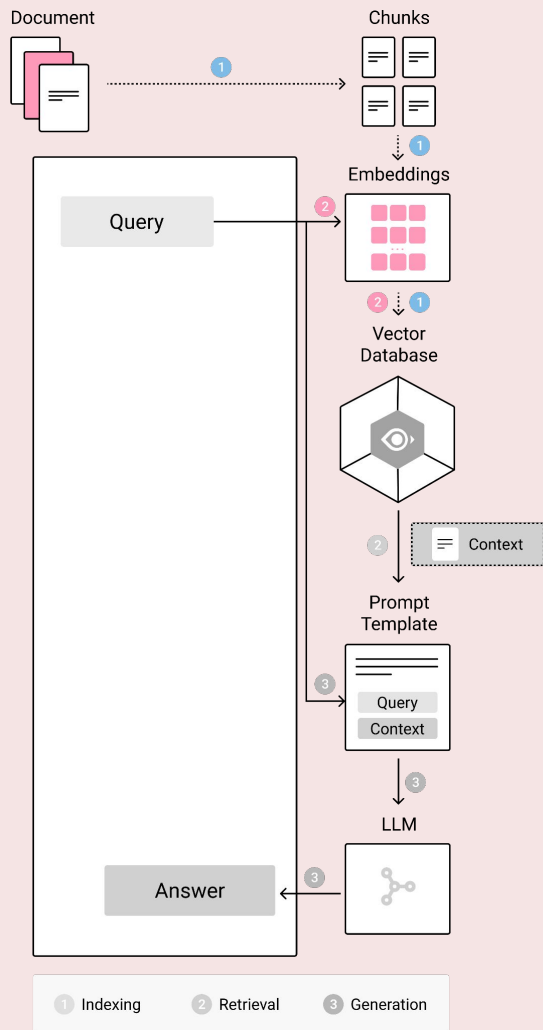
CHUNKING



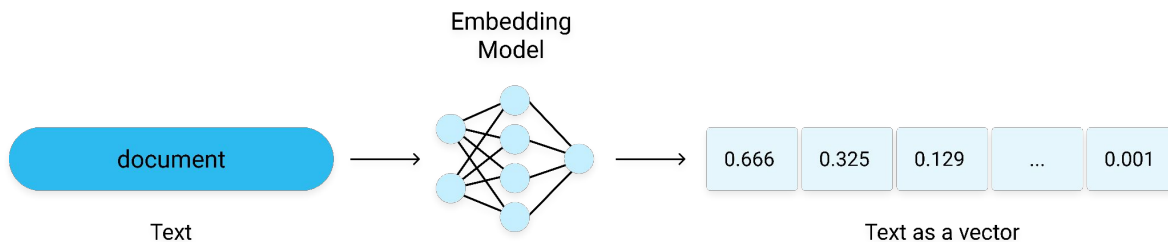


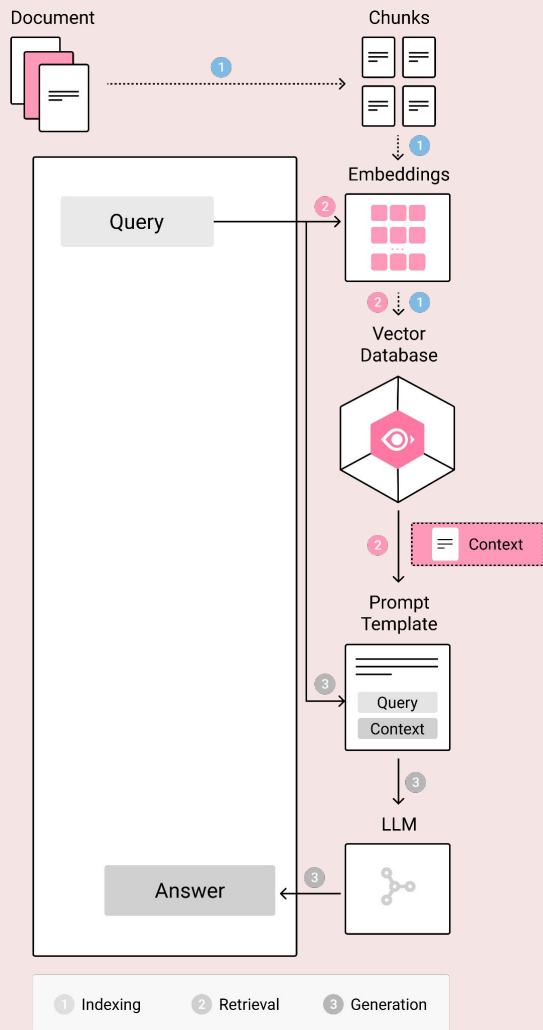
EMBEDDING



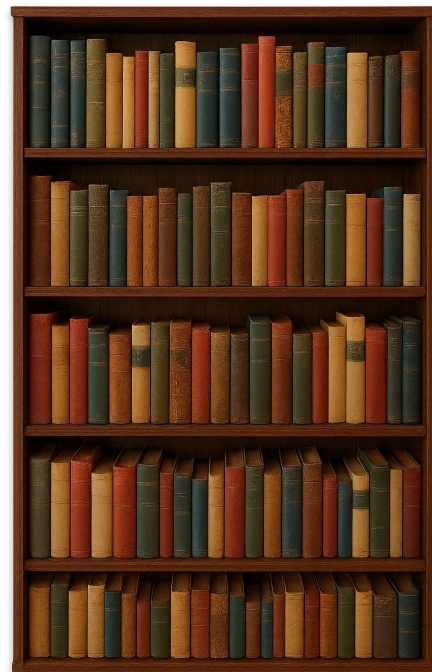


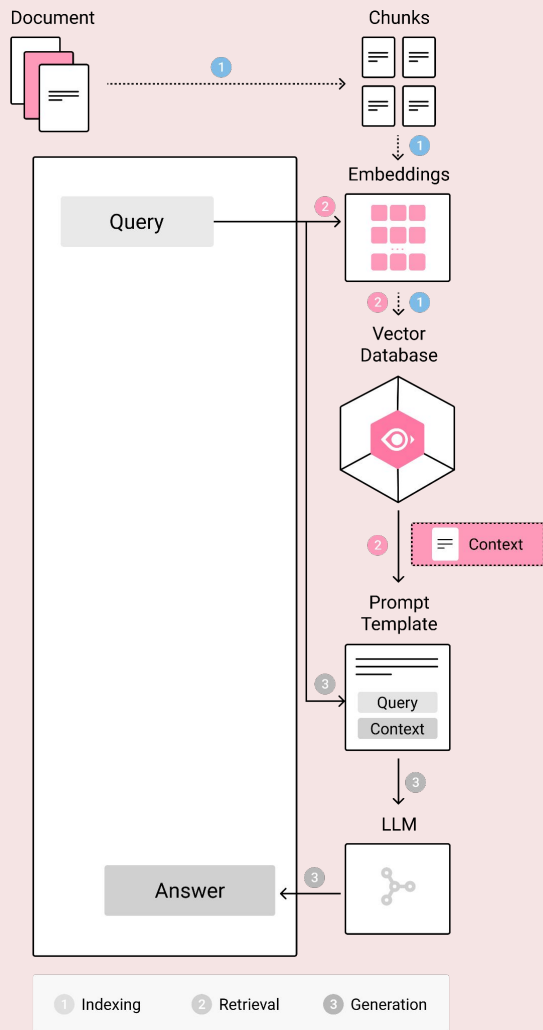
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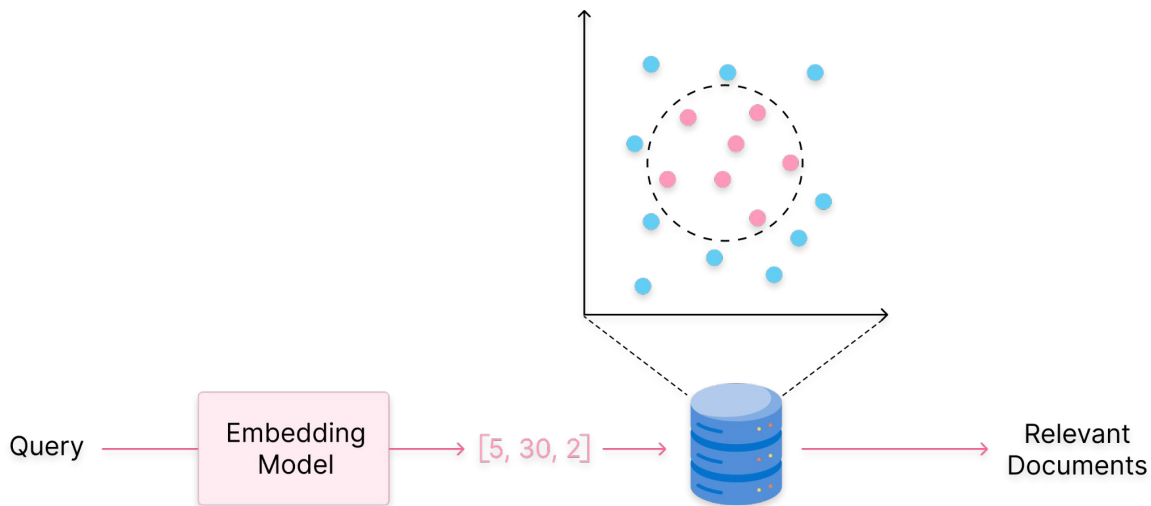


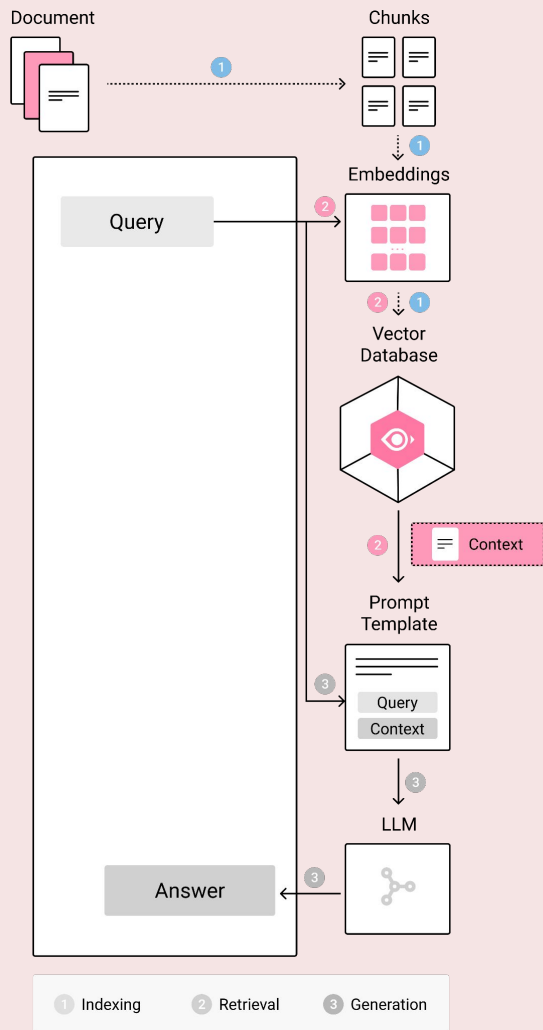
VECTOR DATABASE



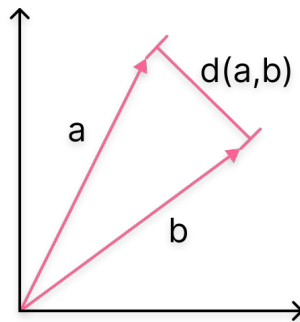


VECTOR DATABASE

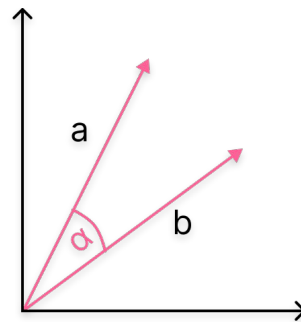




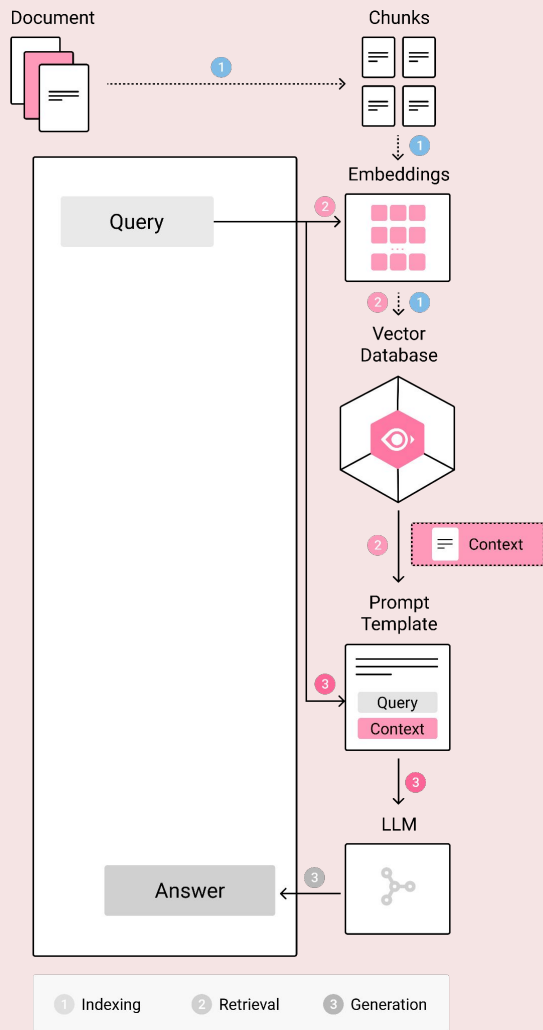
VECTOR DATABASE



Euclidean Distance



Cosine Similarity



PROMPT TEMPLATE

You're a friendly customer support. You know this:

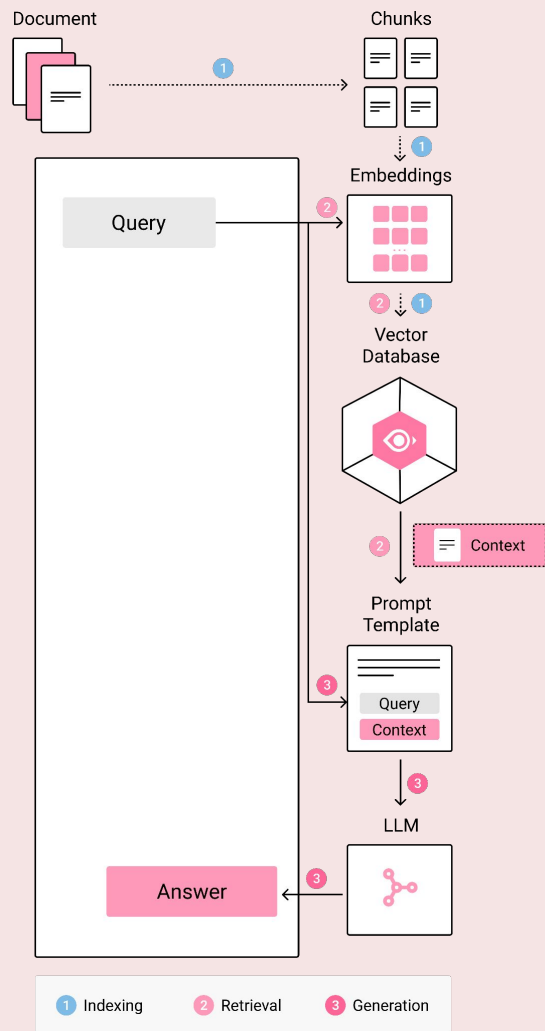
<EXTERNAL KNOWLEDGE>

A customer asked this:

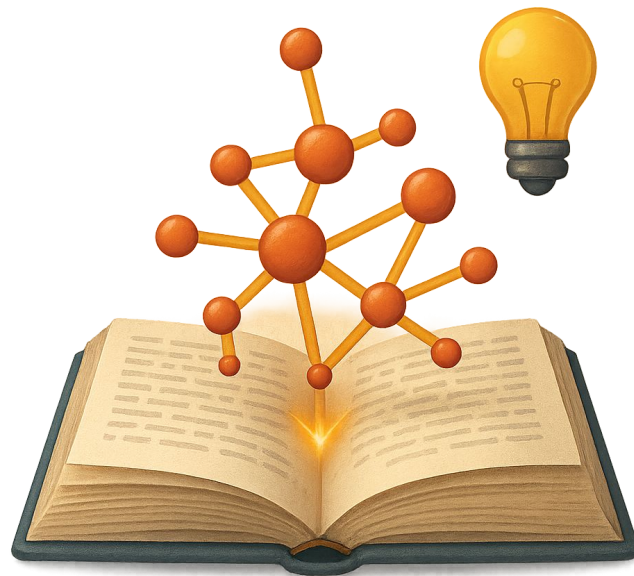
<USER QUERY>

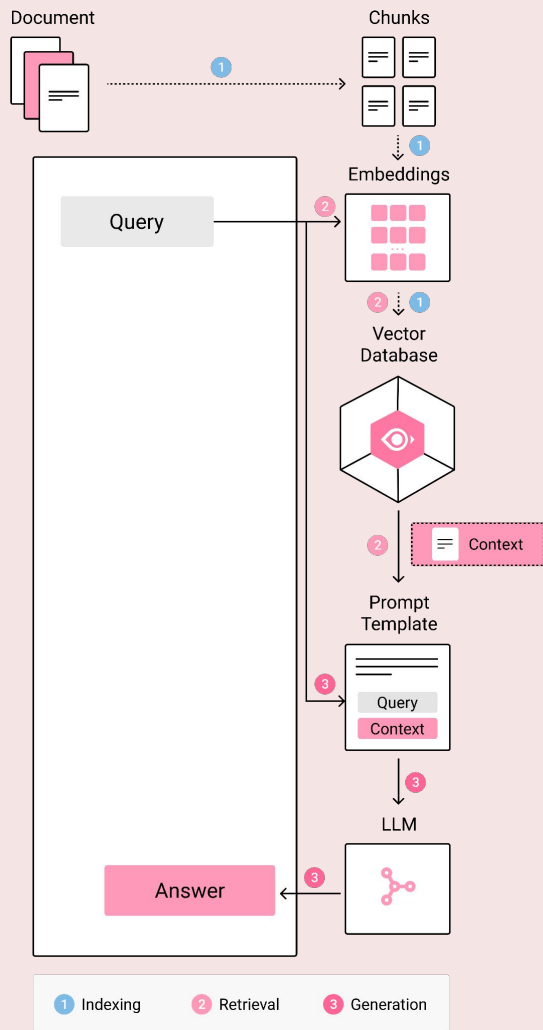
Please generate an answer for the customer.





LARGE LANGUAGE MODEL





LARGE LANGUAGE MODEL



Hugging Face

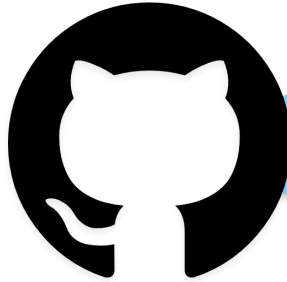


Ollama

OpenAI

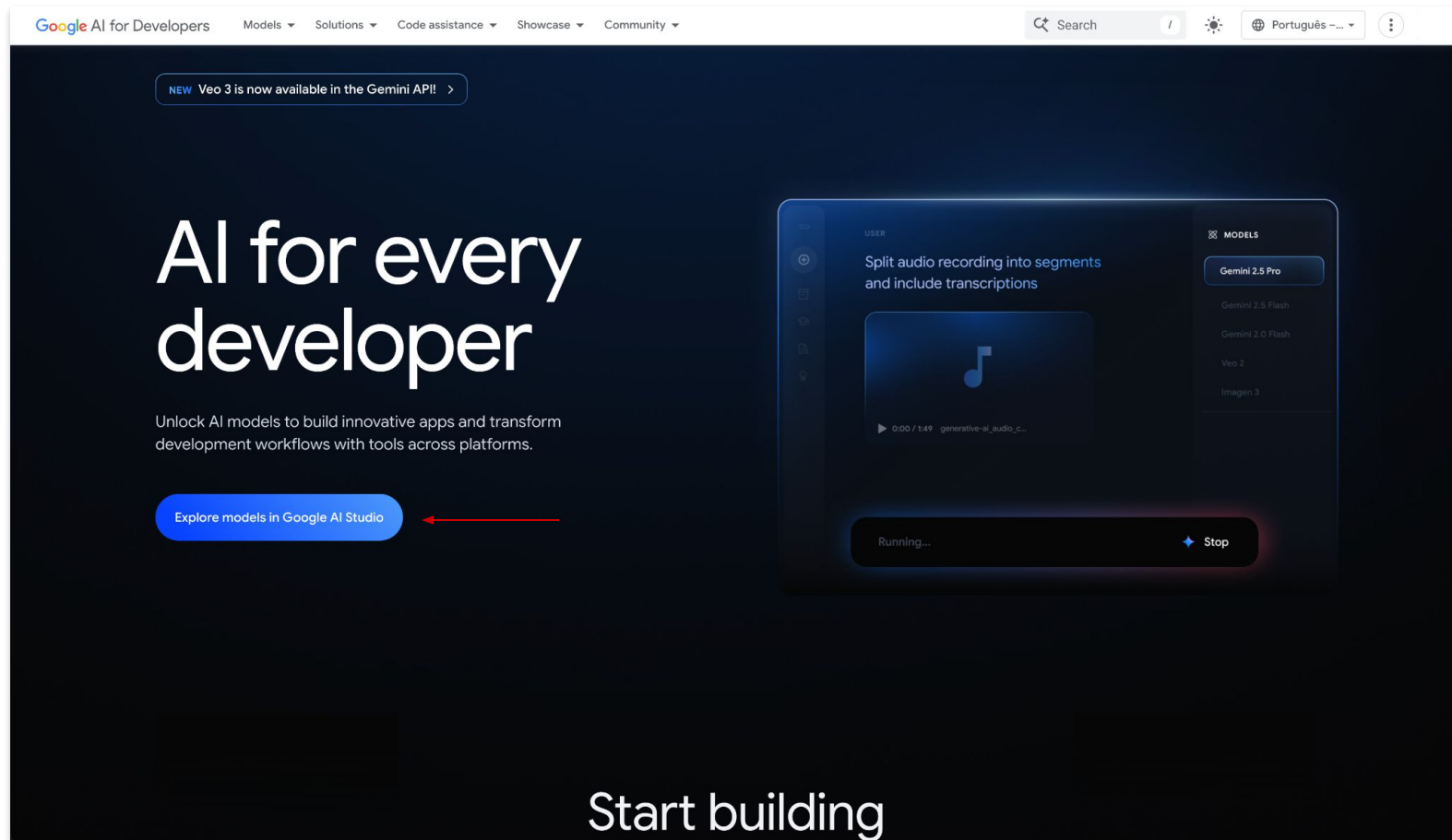
Gemini

REPOSITORY



<https://github.com/conect2ai/2025-UNIBS-Courses/>

TUTORIAL



The screenshot shows the Google AI for Developers homepage. At the top, there's a navigation bar with links for Models, Solutions, Code assistance, Showcase, and Community. A search bar and language selector (Português) are also present. A banner at the top left announces 'NEW Veo 3 is now available in the Gemini API!'. The main heading 'AI for every developer' is prominently displayed. Below it, a subheading states 'Unlock AI models to build innovative apps and transform development workflows with tools across platforms.' A blue button labeled 'Explore models in Google AI Studio' is highlighted with a red arrow. To the right, a preview of the Google AI Studio interface is shown, featuring a chat window with a user prompt 'Split audio recording into segments and include transcriptions', a media player with a music note icon, and a sidebar with a list of models including Gemini 2.5 Pro, Gemini 2.5 Flash, Gemini 2.0 Flash, Veo 2, and Imagen 3. At the bottom of the preview, a 'Running...' status bar with a 'Stop' button is visible.

NEW Veo 3 is now available in the Gemini API! >

AI for every developer

Unlock AI models to build innovative apps and transform development workflows with tools across platforms.

Explore models in Google AI Studio

USER

Split audio recording into segments and include transcriptions

MODELS

- Gemini 2.5 Pro
- Gemini 2.5 Flash
- Gemini 2.0 Flash
- Veo 2
- Imagen 3

Running... Stop

Start building

https://ai.google.dev/

Google AI Studio

Chat

Stream

Generate media

Build

History

Enable saving

Chat prompt

Generate Python code for a simple calculator app →

Run

What's new



URL context tool
Fetch information from web links



Native speech generation
Generate high quality text to speech with Gemini



Live audio-to-audio dialog
Try Gemini's natural, real-time dialog with audio and video inputs



Native image generation
Interleaved text-and-image generation with Gemini 2.0 Flash

Google AI models may make mistakes, so double-check outputs.

Get API key

Studio Dashboard Documentation

Run settings

Gemini 2.5 Pro

Token count 0 / 1,048,576

Temperature 1

Media resolution Default

Thinking

Thinking mode

Set thinking budget

Tools

Structured output

Code execution

Function calling

Grounding with Google Search

URL context

Advanced settings

Google AI Studio

API keys

Usage & Billing

Changelog

View status

Get API key

Studio

Dashboard

Documentation

Chaves de API

Testar rapidamente a API Gemini

[Guia de início rápido da API](#)

```
<> Code
curl "https://generativelanguage.googleapis.com/v1beta/models/gemini-2.0-flash:generateContent" \
-H 'Content-Type: application/json' \
-H 'X-goog-api-key: GEMINI_API_KEY' \
-X POST \
-d '{
  "contents": [
    {
      "parts": [
        {
          "text": "Explain how AI works in a few words"
        }
      ]
    }
  ]
}'
```

Suas chaves de API estão listadas abaixo. Também é possível visualizar e gerenciar seu projeto e as chaves de API no Google Cloud.

Look up API Key for project

Número do projeto	Nome do projeto	Chave de API	Criação	Plano
Crie uma chave de API para ver seus projetos				

Use as chaves de API com segurança. Não as compartilhe nem as incorpore em códigos públicos. O uso da API Gemini em um projeto com faturamento ativado está sujeito aos [preços do modelo de pagamento por uso](#).

Google AI Studio

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+ Criar chave de API

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    }
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```

Criar chave de API

Selecione um dos seus projetos atuais do Google Cloud

Search Google Cloud projects

⇒ Criar uma chave de API em um projeto atual

Suas chaves de API estão listadas abaixo. Também é possível visualizar e gerenciar seu projeto e as chaves de API no Google Cloud.

Look up API Key for project

Nome do projeto	Nome do projeto	Chave de API	Criação	Plano
Crie uma chave de API para ver seus projetos				

View status

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      ]
    }
  ]
}'
```

Criar chave de API

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Search Google Cloud projects

 Gemini API gen-lang-client-0581720549

Suas chaves de API estão listadas abaixo. Também é possível visualizar e gerenciar seu projeto e as chaves de API no Google Cloud.

Look up API Key for project

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View status

Google AI Studio

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Usage & Billing

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[Guia de início rápido da API](#)

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      "parts": [
        {
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      ]
    }
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}'
```

Criar chave de API

Selecione um dos seus projetos atuais do Google Cloud

Search Google Cloud projects

Q Gemini API (gen-lang-client-0581720549)

Criar uma chave de API em um projeto atual

Suas chaves de API estão listadas abaixo. Também é possível visualizar e gerenciar seu projeto e as chaves de API no Google Cloud.

Look up API Key for project

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View status

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Guia de início rápido da API

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-H "Content-Type: application/json" \
-H "X-goog-api-key: GEMINI_API_KEY" \
-X POST \
-d '{
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    {
      "parts": [
        {
          "text": "E"
        }
      ]
    }
  ]
}'
```

Chave de API gerada

Use suas chaves de API com segurança. Não compartilhe nem as incorpore em códigos que o público possa ver.

AlzaSyAMaONxG9FojqVGFNj9X3T9_pv-vzErmTE

Copiar

Suas chaves de API estão listadas abaixo. Também é possível visualizar e gerenciar seu projeto e as chaves de API no Google Cloud.

Look up API Key for project

Número do projeto	Nome do projeto	Chave de API	Criação	Plano
...1456	Gemini API	...rmTE	25 de ago. de 2025	Configurar o faturamento Ver dados de uso

Use as chaves de API com segurança. Não as compartilhe nem as incorpore em códigos públicos. O uso da API Gemini em um projeto com faturamento ativado está sujeito aos [preços do modelo de pagamento por uso](#).

HANDS-ON